Major Project Report

on

Deep Learning for classification and identification of Bols and Taals of the Tabla in Hindustani Classical music

Abstract

The Tabla, a key percussion instrument in North Indian classical music, is based on specific Taals (rhythmic patterns) and Bols (mnemonic syllables). Accurately identifying and transcribing these elements is a complex task requiring deep expertise in Hindustani music. This paper introduces a deep learning-based approach to automate the detection of Taals and Bols. The study develops curated datasets for both, and tests five deep learning models (Sequential, CNN, RNN, DNN, and SVM) to assess their performance. The results show that the Sequential model performs best, achieving 97% accuracy for Taal detection and 91% for Bol detection. This demonstrates the potential of deep learning for automatic Tabla rhythm recognition. The datasets developed are made publicly available to support reproducibility and future research in this area.

1. Introduction

Recent advancements in audio signal processing due to machine learning and deep learning have led to several innovative applications across a wide spectrum, from speech recognition and synthesis to music generation and noise reduction. Deep learning models such as WaveNet (Oord, et al. 2016) and end-to-end systems such as Deep Speech (Hannun, et al. 2014) have significantly improved the accuracy and naturalness of speech processing. Generative models like MuseNet (Sexton 2023) and source separation tools like Spleeter (Romain, et al. 2020) have transformed music creation and production. Convolutional Neural Networks (CNNs) and Long Short-term Memory (LSTM)-based models are known to be effective for emotion recognition in speech and are being incorporated into virtual assistants and mental health monitoring systems. Neural audio codecs such as Lyra and SoundStream (Zeghidour, et al. 2022) have led to more efficient audio compression; adaptive audio systems have created personalized sound profiles for enhanced listening experiences. ML-based personalized sound profiles and adaptive audio systems are helping to personalize the audio experience of each individual based on their preferences.

Machine learning and deep learning models are increasingly being applied to evaluate and comprehend the complex patterns and structures present in music signals, leading to several innovations in the realm of audio data applications in recent years. Roy et. al. (Roy, et al. 2022) propose a dilated CNN model to classify urban sound by carrying out feature extraction and data augmentation techniques. Environmental sounds are classified based on spectrograms using deep learning networks (Khamparia A. , Gupta D. , Nguyen N. G., Khanna A., Pandey B. and Tiwari P. 2019). A comparative study of the effectiveness of various machine learning techniques in the context of acoustic bird detection is presented by Stowell et. al. (Stowell, et al. 2019).

The interaction of musicology and technology has also led to significant advancements in the field of audio signal processing and machine learning in recent years, particularly in the domain of recognition and classification of musical instruments and notes. Several studies have investigated the application of machine learning and deep learning algorithms for the classification and recognition of musical instruments c.f. (Sreekar and Reddy 2022), (Anuz, et al. 2021), (Blaszke and Kostek 2022), and (Agostini, Longari and Pollastri 2003). Ke Xu (Xu 2021) explores the application of deep neural networks for music genre classification, using attention mechanisms and recurrent neural networks (RNNs) to design a classification model focusing on Chinese traditional musical instruments. Chen et. al. (Chen, et al. 2019) introduce

a novel two-step approach for music genre classification, termed TSMGC, leveraging the analytic hierarchy process (AHP) to weigh musical features. Han, Kim and Lee (Han, Kim and Lee 2017) propose a CNN framework for recognizing the predominant instrument in real-world polyphonic music using two different aggregation methods, aiming to minimize the suppression of sporadically appearing instruments during the averaging process. Bosch et. al. (Bosch, et al. 2012) enhance the identification of predominant music instruments in poly-timbral audio by employing various strategies, including sound segregation algorithms and classification models. Guignard and Kehoe (Guignard and Kehoe 2015) develop a classification system for single-instrument recordings across eight different instruments to produce findings and outcomes that can be utilized in the ongoing investigation of separating multiple sources from a single mono-track recording. Similarly, Singh et. al. (Singh, et al. 2019) address the problem of instrument recognition in music tracks, using a combination of CNN and Support Vector Machine (SVM) to classify audio excerpts pre-processed into images.

The Tabla is a percussion instrument from the Indian subcontinent, used in traditional, classical, popular, and folk music. It consists of two drums of different sizes and timbres: the smaller drum (called the dayan or tabla) is usually played with the dominant hand and produces a higher pitch, while the larger drum (called the bayan or duggi) is played with the other hand and produces a deeper sound. A typical Indian Tabla is shown in Fig. 1. In the context of Hindustani (North Indian) Classical music, the term *taal* refers to the rhythmic cycle or pattern that provides a framework of beats (*matras*) divided into sections (*vibhags*) and represented by a basic pattern called *theka*. Popular taals include Teen taal (16 beats), Ek taal (12 beats), Jhap taal (10 beats), and Rupak taal (7 beats). The term *bol* refers to the mnemonic syllables or sounds that represent the various strokes and patterns played on the tabla, such as "Dha," "Dhin," "Na," "Tin," and "Ta," each with a distinct sound and technique. While a taal provides the overall rhythmic structure, it is the bols that are the building blocks used to articulate this structure on the tabla. A tabla player uses bols to play the theka of a particular taal, maintaining the rhythmic cycle throughout a performance, thus making taal and bol fundamental to the rhythmic expression in tabla rendition.

Various research efforts focus on the analysis, grouping and transcription of Tabla performances, using computational and machine learning techniques. Kshirsagar et al. (Raghavendra, Kshirsagar and Raval 2020) propose a visual-based algorithm for automated transcription of tabla syllables, highlighting the importance of visual cues in understanding tabla playing techniques and syllable transcription. Chordia et al. present a variable-length Markov model-based framework for predicting Tabla sequences (Chordia, Sastry and Şentürk

2011). Gillet and Richard (Gillet and Richard 2004) present a statistical machine learning approach for transcribing tabla signals, aiming to automatically label tabla audio signals. Gowriprasad et al. (Gowriprasad, Aravind and Murthy 2022) present novel SVM- and CNN-based approaches for segmenting solo Tabla concerts into meaningful sections and identifying the associated Tabla *gharana* (stylistic school). While a significant amount of extant research relates to the identification, prediction, and segmentation of tabla performances using various machine learning and statistical methods, no work has been done thus far on the classification of taals and bols for the Tabla in Hindustani classical music.

Automating the task of Tabla taal and bol recognition in Hindustani classical music can help immensely in documenting and evaluating tabla performances. It can also facilitate interactive learning tools and virtual accompaniment systems, fostering practice and performance improvements. However, instruments like the Tabla – known for the complex rhythms and strokes it has been used to produce over the years – introduce unique challenges and opportunities for developing algorithms capable of analysing and understanding these complex musical structures as it is a percussion instrument.

This work presents, to the best of the authors' knowledge, the first attempt at applying deep learning algorithms and methodologies for developing a system capable of accurately detecting and classifying tabla bols (strokes) and taals (rhythms) from audio recordings of the Hindustani classical Tabla. The novel contributions made in this work are as follows. Two enhanced datasets, one each of Taals and Bols of the Tabla are created. Each dataset consists of eight classes and 100 audio files per class, resulting in totally 800 audio files per dataset. The relative effectiveness of five machine learning/ deep learning classifiers, namely Sequential, Convolutional Neural Network (CNN), Recurrent Neural Network (RNN), Deep Neural Network (DNN) and Support Vector Machine (SVM) for classification and Tabla note identification is presented on the Taal and Bol datasets. Such classifiers can be utilized for practical applications such as automatic transcription of Tabla Bols and Taals, as well as for the recognition of Bols and Taals during live performances. All datasets developed as part of this work have been made openly available[1], with the purpose of fostering further research in this area and related fields.

[1] https://github.com/TarangSingh03/Tabla-Taal-and-Bol-Identification



Fig 1: The Tabla, used for accompaniment in Indian (Hindustani) Classical Music

2. Material and Methods

Several state-of-the-art deep learning models are implemented and evaluated for Bol (musical stroke) and Taal (rhythm) classification. For this purpose, audio data is converted in coefficients and image form i.e., MFCCs and Mel spectrograms respectively. Thereafter, the classification algorithms are applied on the datasets and their feature subsets. The workflow of the experimental work is summarized in Fig. 2.

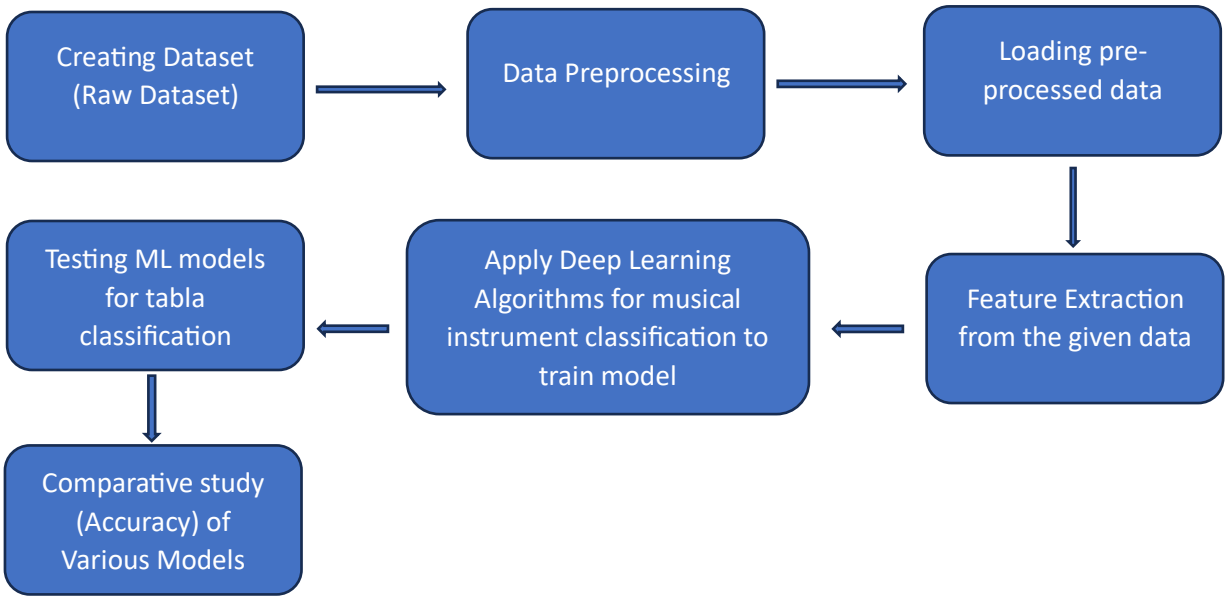


Fig 2: Methodology used to create deep learning model for classifying Bols and Taals of Tabla

The first step in the workflow involved importing and preprocessing the datasets. This included cleaning the audio, trimming, and converting it into MFCC coefficients. Thereafter, a quick visualization was performed with Mel-spectrograms. The data was then divided into training and testing sets, and the various deep learning models were trained, and subsequently tested to assess their accuracy on the Bol and Taal datasets. The key elements of the methodology adopted are described in detail as follows:

2.1 Datasets

Two datasets were created as part of the work. The first of these is a Tabla Taal dataset of eight taals, consisting of 100 files each, resulting in a total of 800 audio files (as shown in Table 1). This dataset was created by generating 239 audio (wav) recordings of the various taals on a tabla, and combining with an existing dataset given by Sumeet Kulkarni (Sumeet n.d.). A separate Bol Dataset was created by the authors and comprises of 800 audio files of eight bols, consisting of 100 files each as shown in Table 2).

S.No.	TAAL	Number of Audio Files
1.	Addhatrital	#100
2.	Bhajani	#100
3.	Dadra	#100
4.	Deepchandi	#100
5.	Ektaal	#100
6.	Jhaptaal	#100
7.	Rupak	#100
8.	Tritaal	#100

Table 1: Summary of Tabla Taal Dataset

S.No.	BOL	Number of Audio Files
1.	Dha	#100
2.	Dhin	#100
3.	Ga	#100
4.	Ka	#100
5.	T	#100
6.	Ta	#100
7.	Tin	#100
8.	Tit	#100

Table 2: Summary of Tabla Bol Dataset

2.2 Dataset Preprocessing

Audio files were pre-processed to make them suitable for model training and testing by eliminating noise and removing silent portions. For this purpose, audio was first trimmed to remove the silent sections from the audio signal. The average duration of the audio files was thus reduced. As an example, the visual representation of a trimmed waveform is given in Fig 3. The corresponding histogram for duration distribution after trimming the input audio files is given in Fig 4, and the Mel- spectrogram and the spectrogram for the visual representation of the audio waveform are given in Fig 5 and Fig 6 respectively. Post trimming, the average duration of files is approximately 5.62 s.

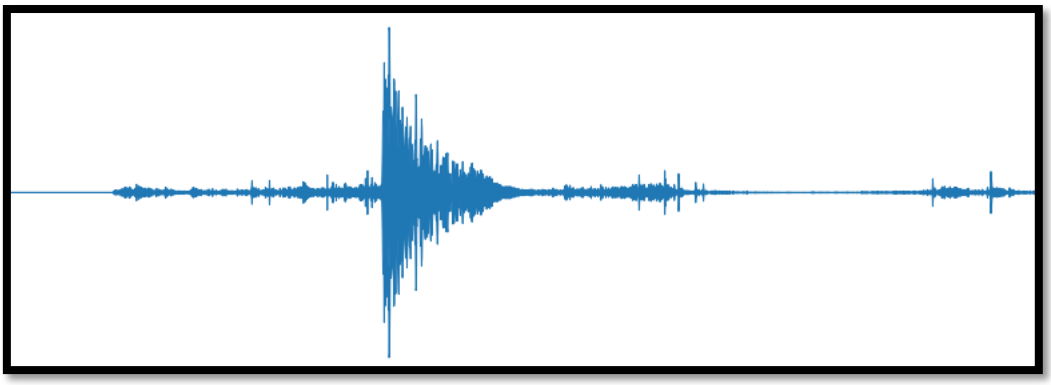


Fig 3: Visual Representation of a Trimmed Waveform

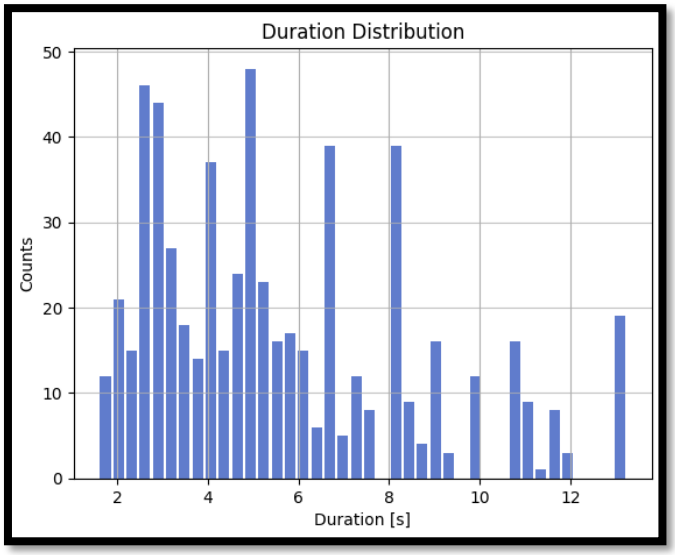


Fig 4: Sample histogram of duration distribution after trimming of audio file



Fig 5: Mel-Spectrogram for visual representation of corresponding audio waveform

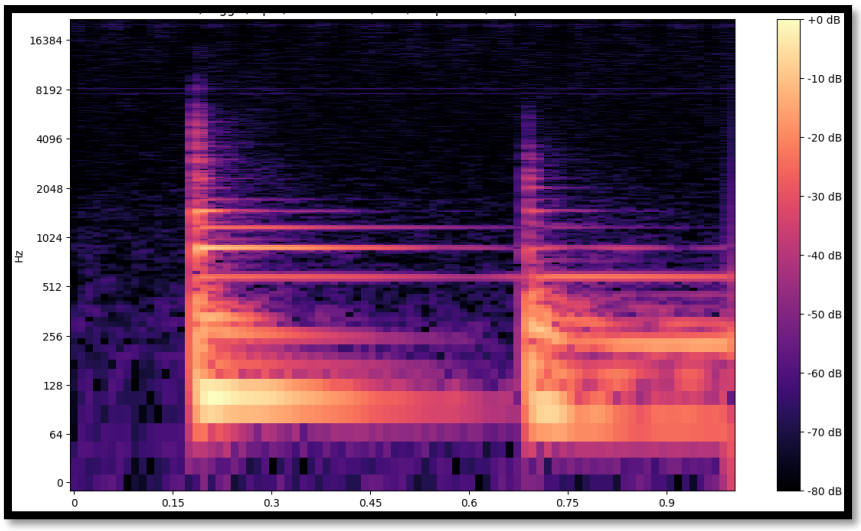


Fig 6: Short-term Fourier Transform Spectrogram of corresponding audio waveform

2.3 Feature Extraction

Features such as RMS (root mean square), chroma_stft, spectral centroid, spectral bandwidth, spectral rolloff, zero-crossing rate, and Mel-frequency cepstral coefficients (MFCCs) were extracted from the audio files. A total of twenty-six features were taken into consideration (as shown in Fig. 7), thirteen of which were MFCCs.

	chroma_stft	rms	spectral_centroid	spectral_bandwidth	\
0	0.475516	0.007484	2069.987213	2309.809583	
1	0.314569	0.008865	1748.547478	2067.108887	
2	0.438885	0.013550	1908.394179	2170.043245	
3	0.341321	0.017542	1768.719729	2097.649678	
4	0.464179	0.017969	2359.998166	2445.914743	
..	...	...	...	...	
795	0.392654	0.022253	2106.756908	2088.726954	
796	0.299215	0.021199	1973.542640	2058.457551	
797	0.376136	0.012936	2518.142425	2447.646233	
798	0.341726	0.015761	2435.470731	2383.170561	
799	0.270703	0.020917	2370.486978	2237.592184	

	rolloff	zero_crossing_rate	mfcc1	mfcc2	mfcc3
0	4360.291148	0.081146	-487.808472	79.516563	2.972535
1	3574.152832	0.083567	-551.521301	71.755371	5.924436
2	3981.437371	0.070801	-427.425873	86.745399	-4.089765

Fig 7: Features extracted from the audio waveforms

2.4 Training and Testing Data

The audio data files were divided into training and testing data, using an 80:20 split. The training set was used to train the model, while testing set was used to verify the results in order to determine the accuracy of the model.

2.5 Model Creation

Four deep learning models, Sequential Model, CNN, DNN and RNN and one machine learning model (SVM) were selected and trained using the training data. While training the models using models.fit() method, we tuned the model by varying values of epochs (number of times training data is passed through the model for training) and batch_size (number of training samples processed before the model's internal parameters (weights) are updated) to obtain the best accuracy. These models are briefly described as follows:

2.5.1 Sequential Model

A deep learning sequential model for musical instrument classification typically involves processing sequential representations of audio data. It involves the use of recurrent neural networks (RNNs) or convolutional neural networks (CNNs) due to the sequential nature of audio data. The sequential model used in this work is a standard feedforward neural network, also known as a multi-layer perceptron (MLP), has three hidden layers, each with 128 neurons, and uses rectified linear unit (ReLU) activation function. The output layer has 8

neurons with SoftMax activation for multi-class classification. The model is designed to process input data with twenty-six features.

2.5.2 Convolutional Neural Network (CNN) Model

The Convolutional Neural Network (CNN) model used in this work analyses spectrogram representations of audio data in order to classify them into different instrumental classes (Bols and Taals). By leveraging convolutional layers, pooling layers, and fully connected layers, the CNN identifies both small-scale and large-scale patterns in the spectrogram data, allowing precise classification of Bols and Taals based on their distinct sound characteristics (features). The proposed CNN (shown in Fig. 8) consists of two Conv1D layers with ReLU activation, followed by MaxPooling1D layers for down-sampling. After flattening the feature maps, it passes through two fully connected layers with SoftMax output layer and ReLU activation for multi-class classification. The input shape is (26, 32), indicating 26 timesteps with 32 features each.

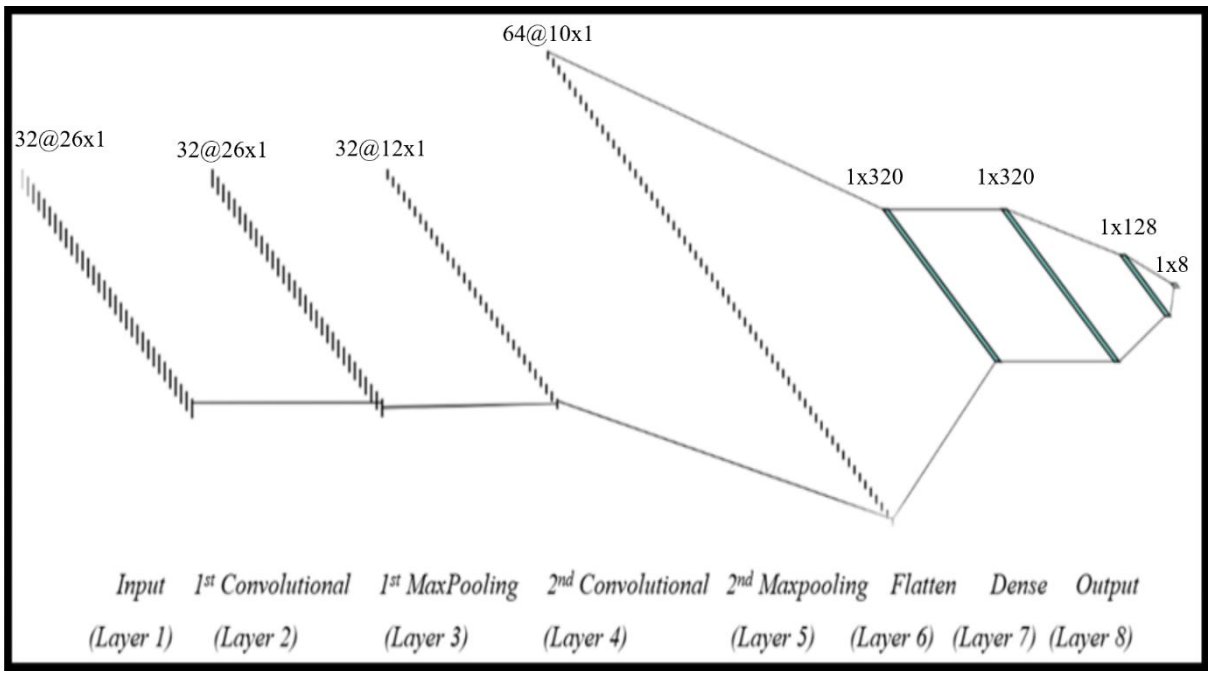


Fig 8: CNN Architecture

2.5.3 Recurrent Neural Network (RNN) Model

Unlike feedforward neural networks that handle each input separately, Recurrent Neural Networks (RNNs) keep track of past information, enabling them to better "understand" how data changes over time. In the proposed RNN model, each time step corresponds to a segment of audio data, such as a frame or window of a spectrogram. The RNN processes these sequential inputs one at a time, updating its internal state at each step. The RNN architecture is shown in Fig. 9 and uses a single recurrent layer object (SimpleRNN layer in

Keras (Chollet 2015)), followed by fully connected layers for classification. The SimpleRNN layer has 128 units and uses ReLU activation, taking input of shape (26, 32), where 26 is the number of time steps and 32 is the number of features at each time step. The output of the SimpleRNN layer is passed through a fully connected layer with 256 units and ReLU activation. Finally, an output layer with 8 units and SoftMax activation for multi-class classification is provided.

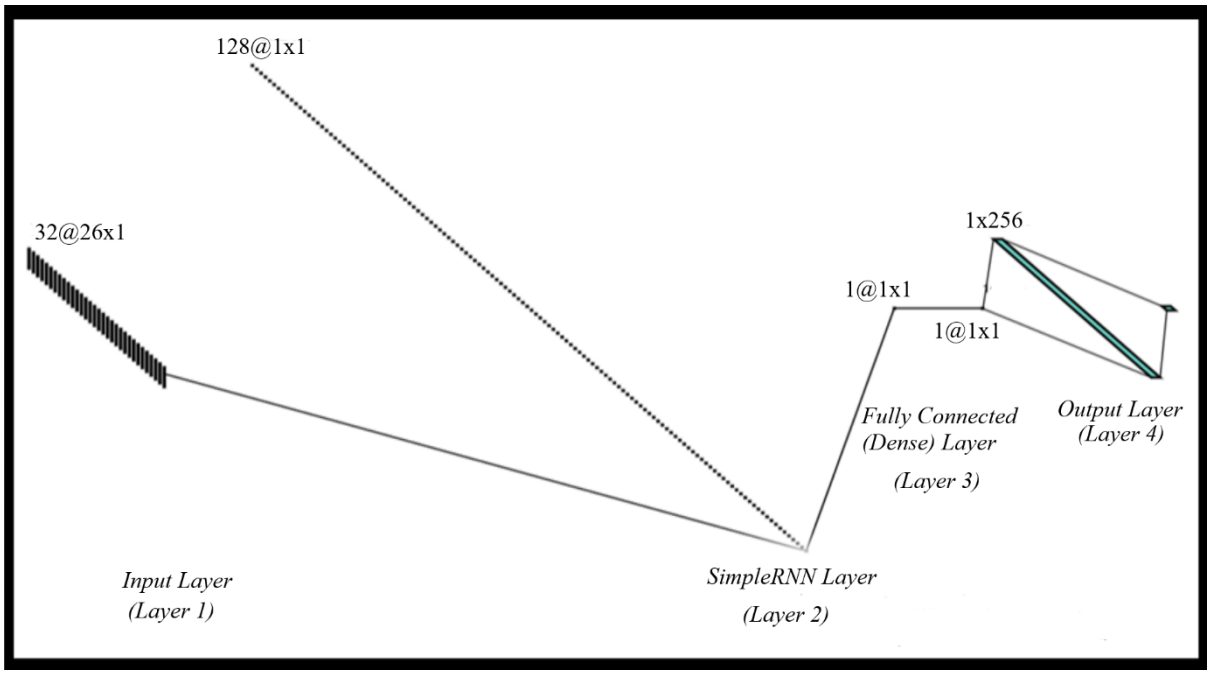


Fig 9: RNN Architecture

2.5.4 Deep Neural Network (DNN) Model

In a DNN model for musical instrument classification, the input data typically consists of features extracted from audio samples, such as spectrograms, Mel-frequency cepstral coefficients (MFCCs), or other relevant representations. These features are fed into the input layer of the DNN, which consists of neurons corresponding to each feature dimension. DNNs consist of multiple hidden layers, allowing them to learn more complex representations of the input data. The proposed DNN uses a simple feedforward neural network, that flattens the input data assuming it has a shape of (26, 32) and converts it into a one-dimensional array. Thereafter two fully connected layers with ReLU activation are applied, comprising 256 and 128 neurons respectively. Lastly, the output layer has 8 neurons and SoftMax activation for multi-class classification, for the 8 classes shown in Fig. 10.

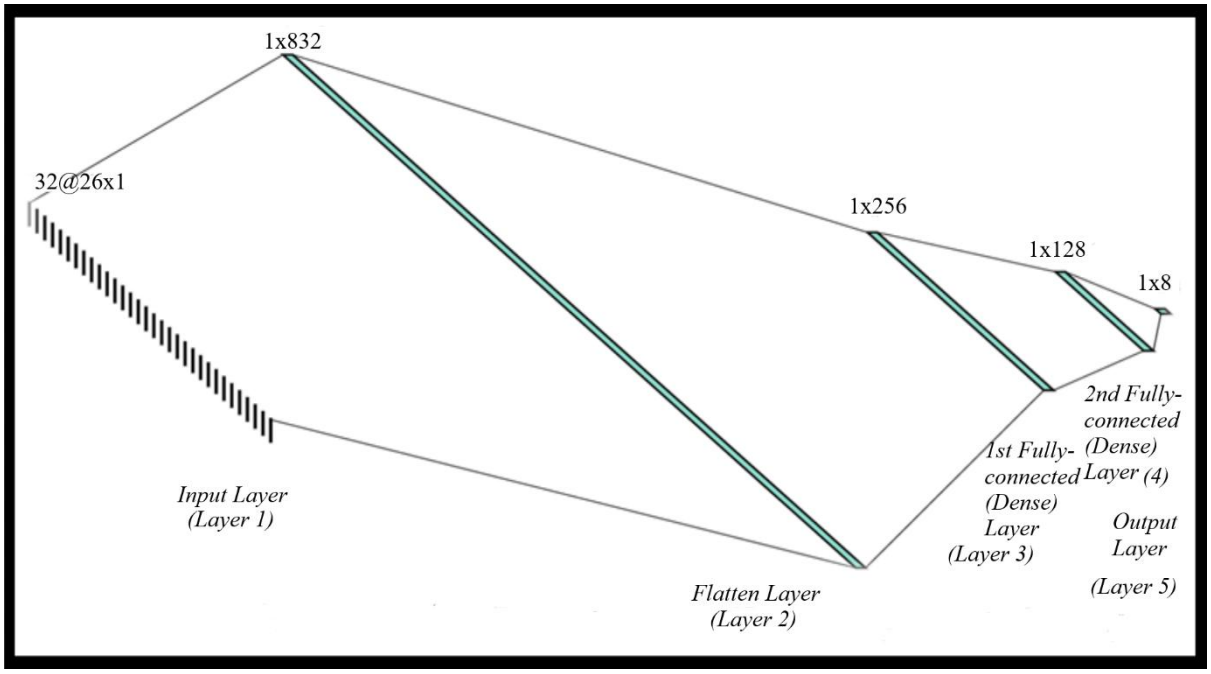


Fig 10: DNN Architecture

2.5.5 Support Vector Machine (SVM) Model

An SVM model for musical instrument classification is a machine learning algorithm designed to classify audio samples into different Bols and Taals based on their acoustic features. A Support Vector Classifier (SVC) with a linear kernel is used, and separates classes using linear decision boundaries, with parameters like penalty term (C), kernel coefficient (gamma), and decision function shape impacting its performance. Unlike the deep learning models discussed above, the SVM focuses on finding the best line to separate different classes in the feature space.

2.6 Training and Testing

The above models were trained and then tested on the Taal and Bol datasets. For training each model, the dataset of extracted features of given audio files was used to determine the relevant features of labelled dataset for determination of different classes or categories of data i.e., different Bols and Taals. After the training phase, each model was tested using the testing data and evaluated using well-known metrics such as accuracy, precision, recall and F1-score. The metrics used are described in the following section.

2.7 Model Evaluation Metrics used

Recall, precision, accuracy, and F1-score are commonly used metrics to evaluate the performance of a classification model. A brief overview of relevant terminology is provided here, followed by a detailed description of each of the evaluation metrics used.

2.7.1 Relevant Terminology:

- *True Positive:* The model correctly classifies (identifies) the Bol or Taal for what it is. For example, when the model correctly classifies a 'Dha' as being a 'Dha'.
- *True Negative:* The model correctly predicts that the given Taal or Bol is NOT some other Taal or Bol. For example, when the model correctly classifies a 'Dha' is NOT being a 'Dhin'.
- *False positive (Type 1 Error):* The model incorrectly classifies a particular Taal or Bol as another Taal or Bol. For example, when the Bol, 'Ta', is wrongly predicted as 'Tin'.
- *False negative (Type 2 Error):* The model incorrectly predicts that a Bol or Taal is not a particular Bol or Taal, when it actually is. For example, a piece of audio contains the Bol 'Dha', but the model predicts that it is not 'Dha'.

These counts are also represented in a confusion matrix that allows us to visualise the performance of each (supervised) machine learning algorithm. Table 3 summarizes these terms in the context of the original Bol in an audio file and the prediction made by the model.

	Original Bol is 'Dha'	**Original Bol is not 'Dha'**
Predicted: 'Dha'	True Positive	False Positive
Predicted: 'Not Dha' or 'is Dhin'	False Negative	True Negative

Table 3: True Positive, True Negative, False Positive and False Negative in the context of multi-class Taal and Bol classification

2.7.2 Recall (Sensitivity):

Recall measures the ability of the model to determine all the Bols (Taals) correctly in the dataset. In other words, it answers the question: "How many were correctly identified by the model among all actual positive instances?" It is also known as sensitivity or true positivity rate.

$$\text{Recall} = \frac{True\ positive}{True\ positive + False\ negative}$$

High recall implies that the model is able to correctly classify most of the Bols (Taals), with minimal false negatives.

2.7.3 Precision:

Precision measures the accuracy of positive predictions made by the model. In the present context, it measures the ability of the model to determine all the Bols (Taals) correctly in the dataset. In other words, it answers the question: " How many are actually positive among all instances predicted as positive?"

$$\text{Precision} = \frac{True\ Positive}{True\ Positive + False\ positive}$$

High precision indicates that the model makes fewer false positive predictions, minimizing the number of incorrect positive predictions.

2.7.4 Accuracy:

Accuracy measures the overall correctness of the model's predictions across all classes. It is calculated as the ratio of the total number of correct predictions (true positives and true negatives) to the total number of predictions made.

$$\text{Accuracy} = \frac{True\ Positive + True\ Negative}{Total\ predictions\ made}$$

Accuracy provides a general understanding of how well the model performs across all Bols and Taals.

2.7.5 F1-score:

The F1-score is the harmonic mean of precision and recall. It provides a balanced measure of both precision and recall, especially when the distribution of Bols and Taals is imbalanced. F1-score ranges from 0 to 1, where a higher value indicates better model performance.

$$\text{F1-score} = 2 \times \frac{precision \times recall}{precision + recall}$$

F1-score is useful when there is an uneven class distribution or when false positives and false negatives have different consequences.

3. Results and Discussion

	Recall	Precision	F1 Score	Accuracy	Ratio of Correct predictions to total number of samples	Best Accuracy Taal/Taals
Sequential Model Evaluation for Taals						
Adhatrital	1.00	1.00	1.00			
Bhajani	0.92	0.92	0.92			
Dadra	0.95	1.00	0.97			
Deepchandi	1.00	1.00	1.00			Deepchandi,
Ektal	1.00	0.87	0.93	0.97	155/160	Adhatrital
Jhaptal	0.91	0.95	0.93			Trital
Rupak	1.00	0.96	0.98			
Trital	0.96	1.00	0.98			
CNN Model Evaluation for Taals						
Adhatrital	0.96	1.00	0.98			
Bhajani	0.85	0.85	0.85			
Dadra	0.93	0.90	0.93			
Deepchandi	0.91	0.95	0.93			Adhatrital
Ektal	1.00	0.87	0.93	0.95	152/160	Rupak
Jhaptal	0.91	1.00	0.95			Trital
Rupak	1.00	1.00	1.00			
Trital	1.00	0.96	0.96			
RNN Model Evaluation for Taals						
Adhatrital	1.00	0.96	0.98			
Bhajani	0.92	0.75	0.83			
Dadra	0.80	0.84	0.82			
Deepchandi	0.87	1.00	0.93			Trital
Ektal	1.00	0.68	0.81	0.85	136/160	Adhatrital
Jhaptal	0.64	0.82	0.72			
Rupak	0.73	0.84	0.78			
Trital	0.91	0.84	0.87			

DNN Model Evaluation for Taals						
Adhatrital	0.96	0.88	0.92			
Bhajani	0.92	0.57	0.71			
Dadra	0.75	1.00	0.86			
Deepchandi	0.70	0.94	0.80			
Ektal	1.00	0.68	0.81	0.87	139/160	Trital
Jhaptal	0.68	1.00	0.81			
Rupak	1.00	0.92	0.96			
Trital	1.00	1.00	1.00			
SVM Model Evaluation for Taals						
Adhatrital	0.93	0.96	0.98			
Bhajani	0.94	0.80	0.85			
Dadra	0.96	1.00	0.93			
Deepchandi	0.93	0.82	0.93	0.92	183/200	Trital
Ektal	0.88	1.00	0.93			
Jhaptal	0.86	0.89	0.95			
Rupak	0.83	0.89	1.00			
Trital	1.00	1.00	0.96			

Table 4: Tabla Taal Dataset Results

The recall, precision and f1-score for each taal, and accuracy with respect of each classification model, and the taal with best performance for a given model are given in Table 4. For the table, it is seen that for the Sequential model, we have 100% recall, precision and f1-score for ‘Adhatrital’ and ‘Deepchandi’, whereas the CNN model obtains 100% recall, precision and f1-score for ‘Rupak Taal’. The RNN model scores 100% recall, 96% precision and 98% f1-score for ‘Adhatrital’ and the Support Vector Machine (SVM) model obtains 100% recall and precision, and 96% f1-score for ‘Trital’. Considering the accuracy of all the models, we can conclude that Sequential is the best model for the Taals of Tabla with overall accuracy of 97%. Also, out of all the taals, ‘Trital’ and ‘Adhatrital; are most accurately predicted by almost all of the models.

Sequential Model Evaluation for Bols						
	Recall	Precision	F1 Score	Accuracy	Ratio of Correct predictions to total number of samples	Best Accuracy Bol/Bols
Dha	0.88	0.95	0.91			
Dhin	0.92	0.80	0.86			
Ga	0.90	1.0	0.95			
Ka	0.96	0.88	0.92	0.91	145/160	Ga
T	0.92	0.80	0.86			
Ta	0.86	0.90	0.88			
Tin	0.86	0.95	0.90			
Tit	0.96	0.92	0.94			
CNN Model Evaluation for Bols						
Dha	0.92	1.00	0.96			
Dhin	1.0	0.81	0.90			
Ga	0.95	0.95	0.95			
Ka	0.96	1.00	0.98			Ga,
T	1.00	0.68	0.81	0.91	145/160	Ka
Ta	0.86	0.86	0.86			
Tin	0.77	0.94	0.85			
Tit	0.87	0.95	0.91			
RNN Model Evaluation for Bols						
Dha	0.79	0.86	0.83			
Dhin	0.69	0.45	0.55			
Ga	0.95	0.86	0.90			
Ka	0.78	0.86	0.82			
T	0.85	0.65	0.73	0.78	124/160	Ga
Ta	0.59	0.87	0.70			
Tin	0.68	0.79	0.73			
Tit	0.87	0.83	0.85			

DNN Model Evaluation for Bols						
Dha	0.92	0.88	0.90			
Dhin	0.92	0.86	0.89			
Ga	1.00	0.91	0.95			
Ka	0.96	0.92	0.94			
T	0.54	1.00	0.70	0.86	137/160	Ga
Ta	0.73	0.76	0.74			
Tin	0.73	0.80	0.76			
Tit	0.96	0.81	0.88			
SVM Model Evaluation for Bols						
Dha	0.9642	0.9000	0.91342			
Dhin	0.8235	0.9333	0.8749			
Ga	1.0000	1.0000	1.0000			
Ka	0.9667	0.9667	0.9667			
T	0.8823	0.8333	0.8571	0.91	181/200	Ga
Ta	0.8571	0.7741	0.8134			
Tin	0.7667	0.9200	0.8363			
Tit	0.9628	0.9285	0.8979			

Table 5: Tabla Bol Dataset Results

Table 5 represents the recall, precision and f1-score for each bol of Tabla and accuracy in respect of each classification model, as also the taal with the best performance for a given model. The best value of recall, precision and f1-score for the Sequential model, i.e. 90% recall, 100% precision and 95% f1-score are observed for the bol ‘Ga’. Similarly, the CNN model gives best performances as 96% recall, 100% precision and 98% f1-score for the bol ‘ka’; the RNN model obtains 95% recall, 86% precision and 90% f1-score for the bol ‘Ga’ and SVM obtains 100% recall, 91% precision and 95% f1-score for the bol ‘Ga’. On comparing the accuracy of all models, we see that the Sequential, CNN & SVM Models obtain the highest accuracy (of 91%) for Tabla Bols. Further, of all the bols, ‘Ga’ was the bol most accurately predicted by all the models.

Deep Learning Model	Taal Dataset Accuracy	Bol Dataset Accuracy
Sequential	97	91
CNN	95	91
RNN	85	78
DNN	87	86
SVM	92	91

Table 6: Accuracy of Deep Learning Model for Taal and Bol Datasets respectively

The overall result and accuracy of each deep learning model with respect to Tabla taal and Bol datasets is summarized in Table 6. The results show that the Sequential Model provides the best accuracy for both Bols and Taals of the Tabla. It can also be observed that the general performance or accuracy of all five classifiers (models) was better for the Tabla Taal dataset, due possibly to the longer duration of the dataset (as each taal consists of multiple bols).

4. Conclusions

This article describes, for the first time, the application of machine learning models for recognition of Bols and Taals of Tabla in the Indian classical (Hindustani) music system. Novel Taal and Bol datasets were developed as part of the work, and the machine learning and deep learning techniques described in the paper were used for training and testing the effectiveness of the proposed models. Various performance evaluation metrics were also used to analyse the performance of the models. The maximum accuracy achieved on a classifier was found to be 97% for Taals and 91% for Bols (Sequential deep learning model). It was observed that among the two datasets, the Tabla Taal dataset was most accurately recognised due to the advantage of long duration of its audio files 'Trital' in Taals and 'Ga' in Bols was most accurately determined by almost all the models. These results highlight the potential of machine learning and deep learning techniques for accurately classifying Tabla Bols and Taals. Future work in this area could also explore the applicability of the proposed models for musical note identification and classification in other musical instruments and genres. The dataset provided in this work could also be used for benchmarking the performance of other models for Tabla Taal and Bol recognition.

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